

BEST Istanbul Yıldız Hackathon 2026

FitSense AI

Your Phone is Your Personal Trainer

- MATLAB Mobile
- Deep Learning
- Signal Processing
- Sensor Fusion
- App Designer

NighQuest | May 2026

THE PROBLEM

**Fitness trackers cost \$200+
but everyone already has a phone**

Smartphones have accelerometers, gyroscopes, and GPS — the same sensors used by expensive wearables. We built an AI-powered system that turns your phone into a professional fitness tracker using MATLAB.



Multi-Sensor Data

Accelerometer + Gyroscope + GPS captured via MATLAB Mobile



AI Classification

Deep Learning LSTM identifies what activity you're doing in real-time



Smart Analytics

Steps, calories, distance, and a unique "Workout DNA" fingerprint

OUR APPROACH

End-to-End Pipeline



Signal Processing

- 4th-order Butterworth bandpass (0.5–15 Hz)
- Zero-phase filtering (filtfilt)
- Gravity removal & magnitude computation
- Adaptive peak detection for steps

Feature Engineering

- 10 time-domain features × 4 channels
- FFT spectral energy & entropy
- Dominant & mean frequency
- Sliding windows (2s, 50% overlap)

SIGNAL PROCESSING

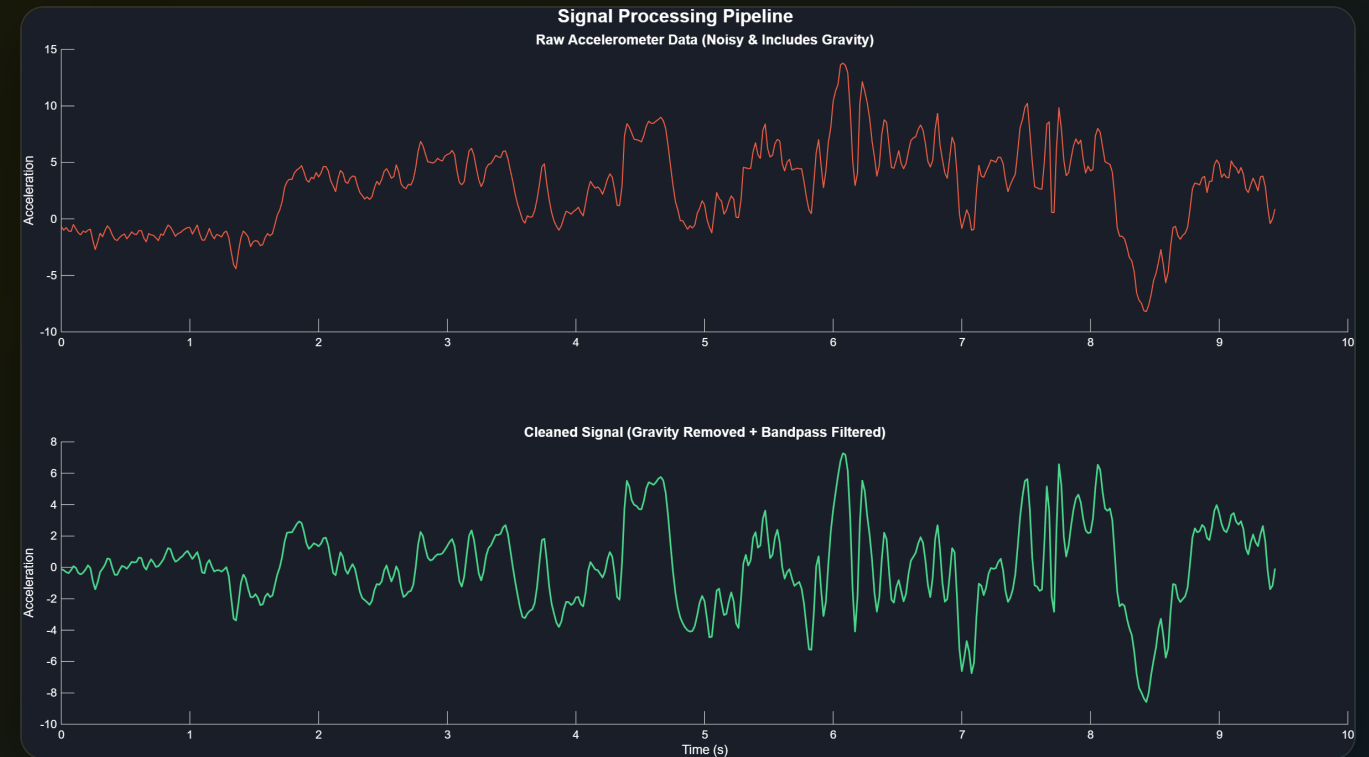
From **Noisy** Data to **Clean** Signals

Raw accelerometer data is noisy and contains gravity. Our preprocessing pipeline:

```
% Butterworth bandpass filter
[b, a] = butter(4, [0.5 15]/(Fs/2));
accelClean = filtfilt(b, a, rawAccel);

% Orientation-independent magnitude
mag = sqrt(ax.^2 + ay.^2 + az.^2);

% Adaptive step detection
[peaks, locs] = findpeaks(mag, ...
    'MinPeakHeight', std(mag)*0.8, ...
    'MinPeakDistance', round(Fs*0.35));
```





- | BiLSTM Activity Classification | | | | | | | | | | |
|--------------------------------|--------------------------|--------------------------|---------|---------|---------|--------|---------|--------|-------|--|
| True Class | fast _w alking | 10 | 1 | | 1 | | | 83.3% | 16.7% | |
| | jogging | | 14 | | | 1 | | 93.3% | 6.7% | |
| | running | | | 9 | | 1 | 1 | 81.8% | 18.2% | |
| | sitting | | | | 13 | | | 100.0% | | |
| | stairs | | | | | 18 | | 100.0% | | |
| | walking | 1 | | | | | 1 | 50.0% | 50.0% | |
| | | 90.9% | 93.3% | 100.0% | 92.9% | 90.0% | 50.0% | | | |
| | | 9.1% | 6.7% | | 7.1% | 10.0% | 50.0% | | | |
| | | fast _w alking | jogging | running | sitting | stairs | walking | | | |
| | | Predicted Class | | | | | | | | |

MODEL COMPARISON

LSTM vs Classic ML

We trained 5 models to compare deep learning against traditional approaches:



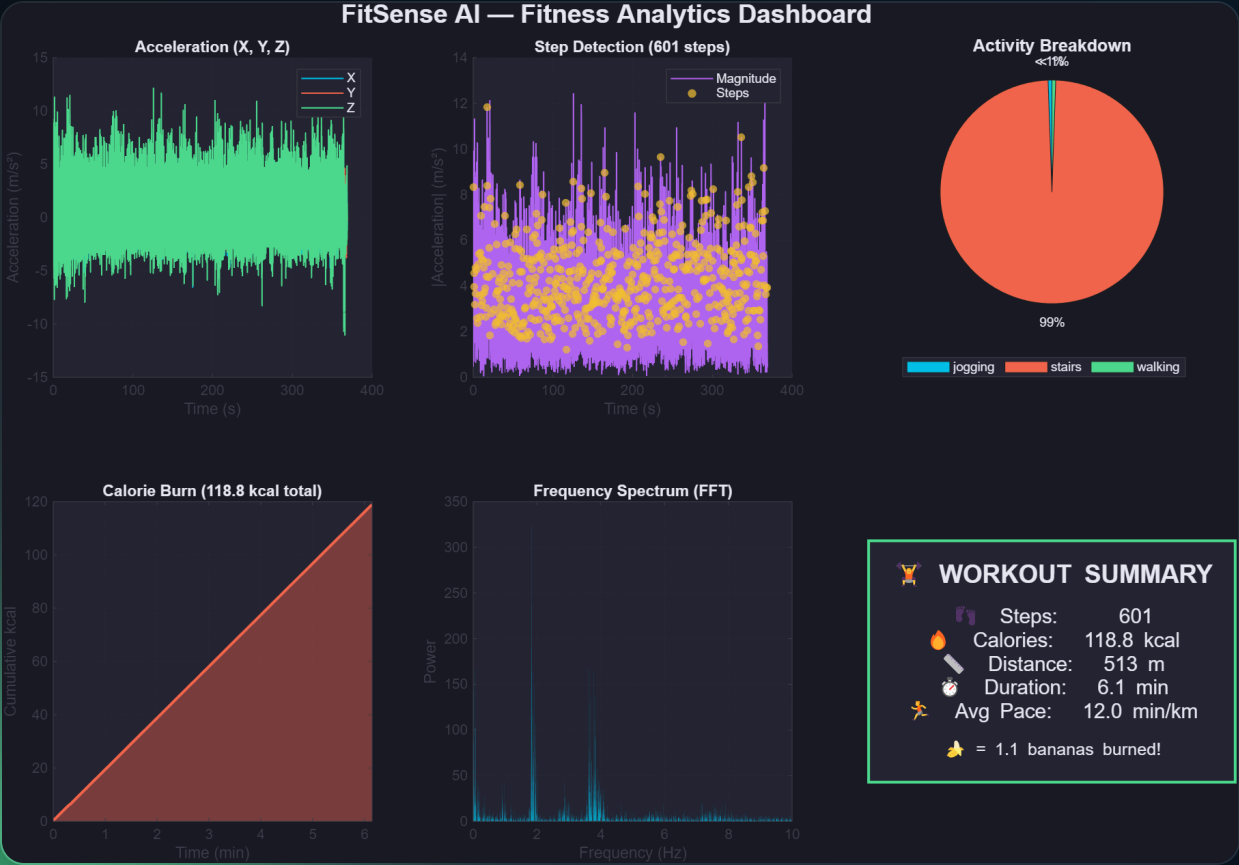
The comparison demonstrates mastery of multiple ML paradigms. The Bidirectional LSTM (BiLSTM) outperformed the classic ML models, proving its ability to capture complex temporal sequence dynamics.

VISUALIZATION

Professional Dashboard & Workout DNA

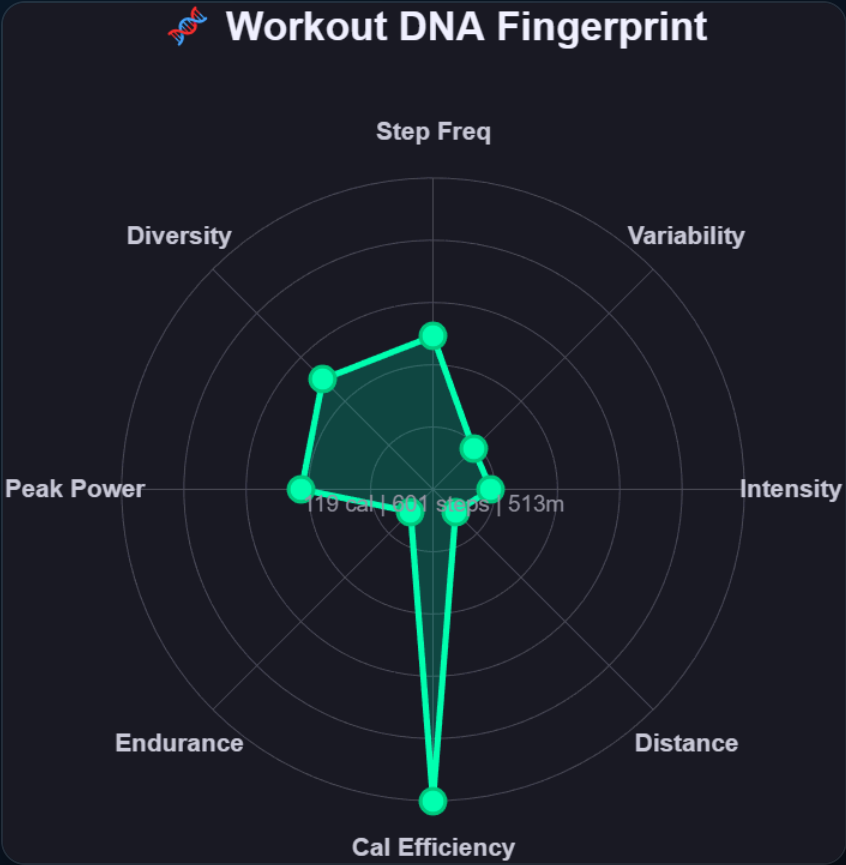
App Designer Dashboard

Interactive GUI with dark theme, tabbed views, real-time results, and user profile input.



Workout DNA Fingerprint

A unique polar radar chart that creates a visual "fingerprint" for each workout based on 8 metrics.



12+ Plot Types

Dark Theme

Interactive Tabs

Animated Plots

3D Trajectory

DEMO

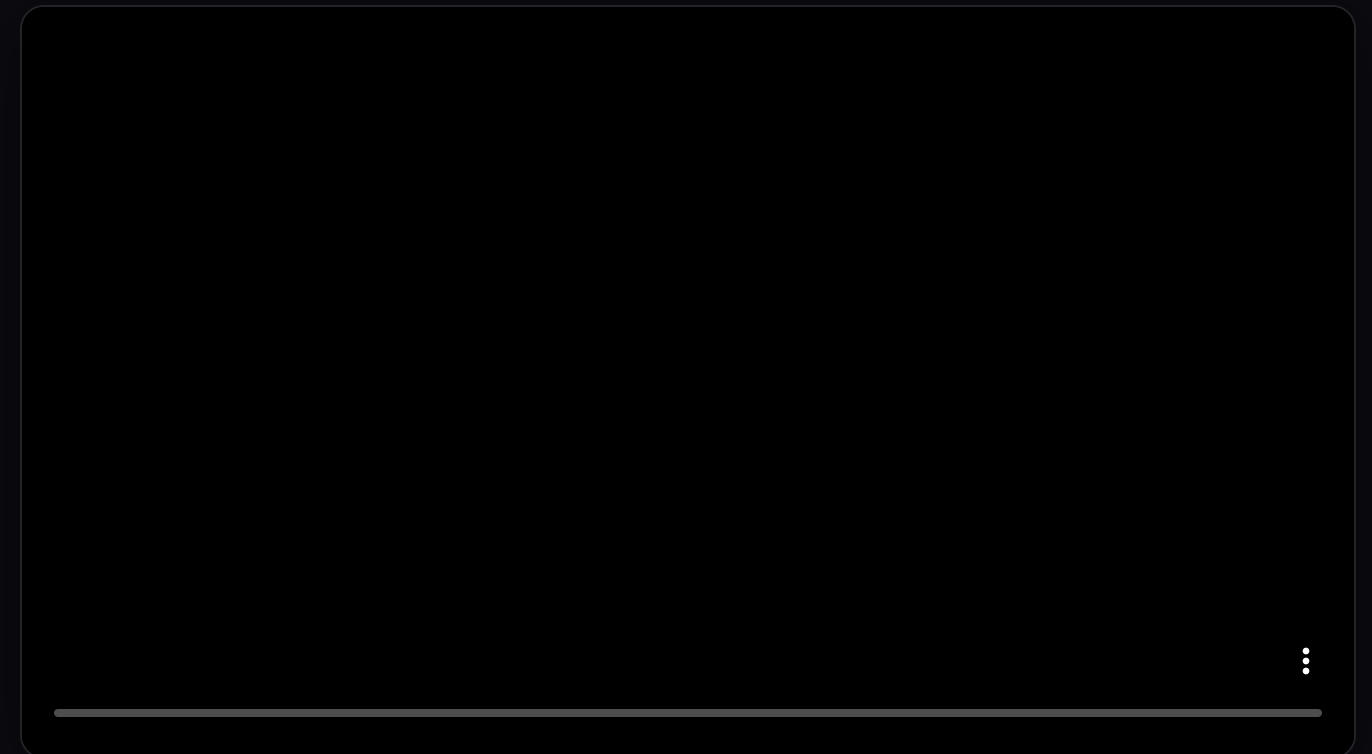
Live Demo & Execution

Watch the FitSense AI app in action, processing real-time sensor data and displaying analytics.

→ Real-time activity recognition

→ Live step counting & calorie estimation

→ Interactive data visualization



RESULTS & IMPACT

Workout Summary

601

Steps Detected

119.5

Calories Burned (kcal)

513

Distance (meters)

6.1

Duration (min)

That's equivalent to...



1.1 bananas



0.4 pizza

slices

Anyone with a smartphone can now have a **professional fitness tracker**

Built with MATLAB Online • MATLAB Mobile • Deep Learning Toolbox • App Designer